

```

import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
from scipy import stats
from scipy.integrate import quad
import seaborn as sns
from scipy import stats

plt.rcParams.update({'font.size': 12})
plt.rcParams['figure.figsize'] = (10, 6)

# Ustawienie seed dla powtarzalności
np.random.seed(42)

# Generowanie danych:  $y \approx 2x + 5 + \text{szum}$ 
n = 50
x = np.linspace(0, 10, n)
y_true = 2 * x + 5 # prawdziwa zależność
noise = np.random.normal(0, 2, n) # losowy szum
y = y_true + noise

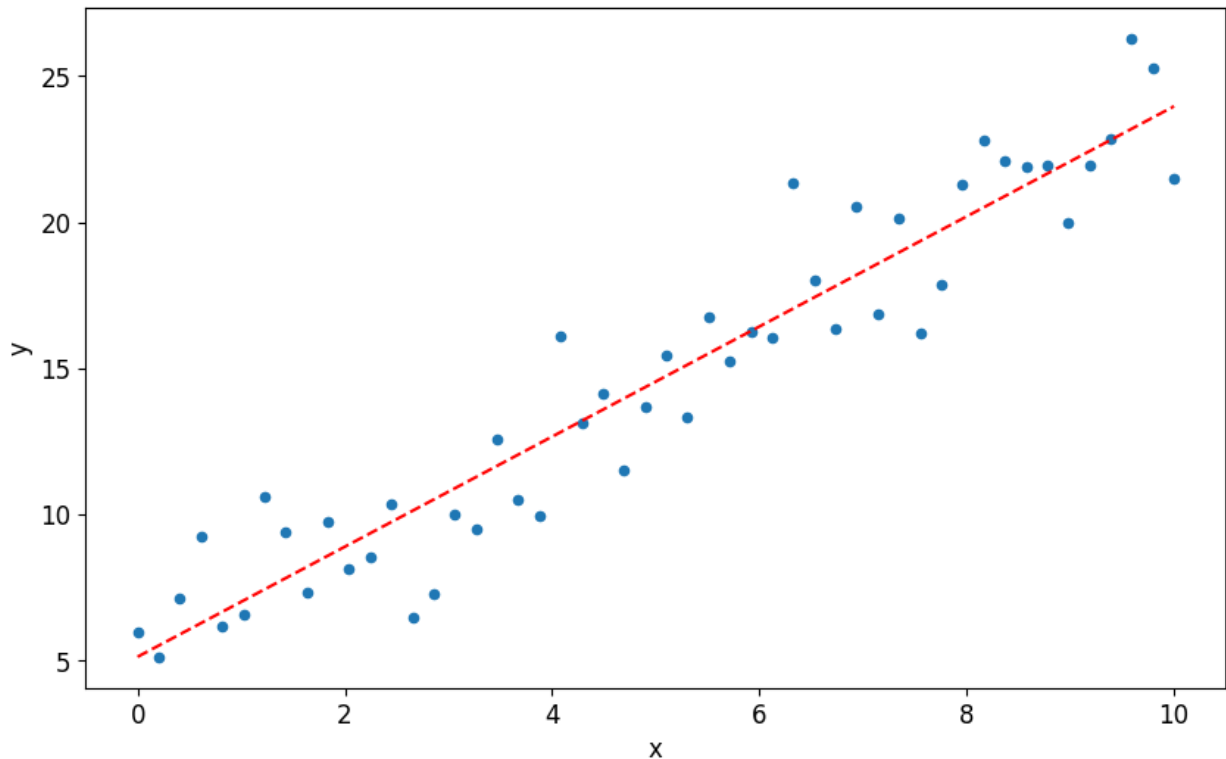
# Stworzenie DataFrame
data = pd.DataFrame({'x': x, 'y': y})

slope, intercept, r_value, p_value, std_err = stats.linregress(data.x,
data.y)
print(f"slope: {slope:.2f}, intercept: {intercept:.2f}")
fig, ax = plt.subplots()
data.plot(kind="scatter", x='x', y='y', ax=ax)
y_linreg = slope*x + intercept
ax.plot(x, y_linreg, 'r--')

slope: 1.88, intercept: 5.13

[<matplotlib.lines.Line2D at 0x7a440dc47380>]

```



```
def fit_and_plot(x, y, ax):
    slope, intercept, r_value, p_value, std_err = stats.linregress(x, y)
    print(f"slope: {slope:.2f}, intercept: {intercept:.2f}")
    print(f"r: {r_value}")
    print(f"err: {std_err}")
    ax.plot(x, y, 'bo')
    x_linspace = np.linspace(min(x), max(x), 30)
    y_linreg = slope*x_linspace + intercept
    ax.plot(x_linspace, y_linreg, 'r-')

np.savetxt("dane_do_fit.txt", np.c_[x, y])

df = pd.read_csv("dane_do_fit.txt", delimiter=" ", header=None,
names=["x", "y"])

df.head()

{"summary": "{\n  \"name\": \"df\",\n  \"rows\": 50,\n  \"fields\": [\n    {\n      \"column\": \"x\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 2.9749754565537248,\n        \"min\": 0.0,\n        \"max\": 10.0,\n        \"num_unique_values\": 50,\n        \"samples\": [\n          2.653061224489796,\n          7.959183673469389,\n          6.122448979591836\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"y\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 5.8977469074465425,\n        \"min\": 5.131634662963752,\n        \"max\": 26.297917921825583,\n
```

```

\ "num_unique_values\ ": 50,\n          \ "samples\ ": [\n
6.479561959663996,\n          21.312089818677023,\n
16.041484734724882\n          ],\n          \ "semantic_type\ ": \ "\",\n
\ "description\ ": \ "\",\n          }\n          }\n          ]\n
n\ },\n "type": "dataframe", "variable_name": "df"}

```

```

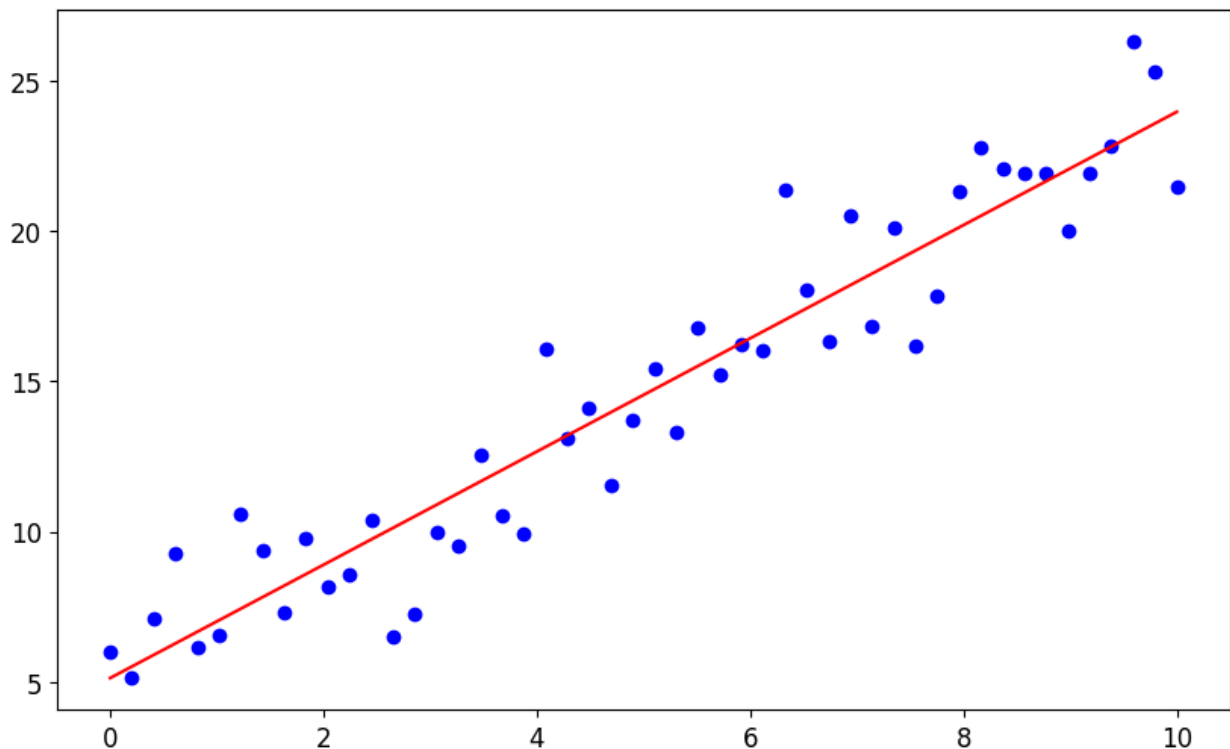
fig, ax = plt.subplots()
fit_and_plot(df['x'], df['y'], ax)

```

```

slope: 1.88, intercept: 5.13
r: 0.9503548762459013
err: 0.08903837899142658

```



Zadanie 1

1. Wczytaj dane:

- Wczytaj dane z pliku `anscombe.csv` do DataFrame za pomocą biblioteki `pandas`.
- Przekształć dane, aby miały format:

	x1	y1	x2	y2	x3	y3	x4	y4
0	10.0	8.04	10.0	9.14	10.0	7.46	8.0	6.58
1	8.0	6.95	8.0	8.14	8.0	6.77	8.0	5.76

2. Stwórz wykresy punktowe:

- Użyj biblioteki `matplotlib` lub `seaborn`, aby stworzyć wykresy.
- Na każdej osi (`ax`) umieść wykres punktowy dla par `x1, y1, x2, y2, x3, y3` oraz `x4, y4`.

```
! head anscombe.csv

0,0,1,1,2,2,3,3
x,y,x,y,x,y,x,y
10.0,8.04,10.0,9.14,10.0,7.46,8.0,6.58
8.0,6.95,8.0,8.14,8.0,6.77,8.0,5.76
13.0,7.58,13.0,8.74,13.0,12.74,8.0,7.71
9.0,8.81,9.0,8.77,9.0,7.11,8.0,8.84
11.0,8.33,11.0,9.26,11.0,7.81,8.0,8.47
14.0,9.96,14.0,8.10,14.0,8.84,8.0,7.04
6.0,7.24,6.0,6.13,6.0,6.08,8.0,5.25
4.0,4.26,4.0,3.10,4.0,5.39,19.0,12.50

import pandas as pd
import matplotlib.pyplot as plt

df_anscombe = pd.read_csv('anscombe.csv', header=None, skiprows=2)
df_anscombe.columns = ['x1', 'y1', 'x2', 'y2', 'x3', 'y3', 'x4', 'y4']

print("Wczytane dane Anscombe:")
print(df_anscombe.head())

fig, axes = plt.subplots(nrows=2, ncols=2, figsize=(10, 8),
sharex=True, sharey=True)
axes = axes.flatten()

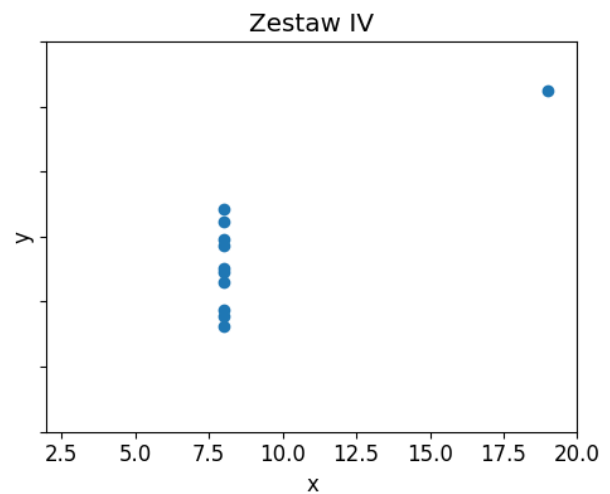
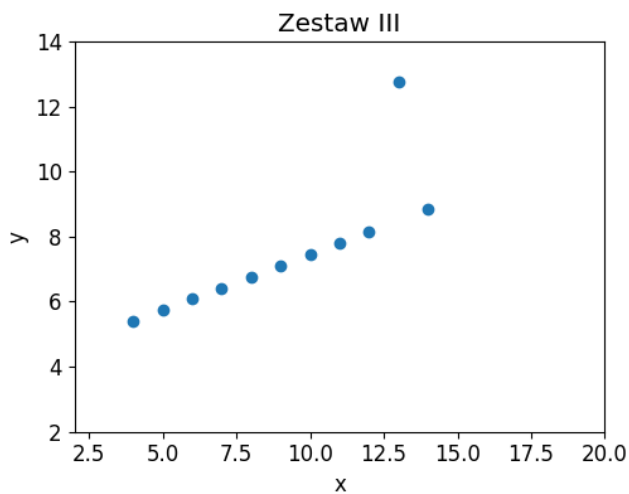
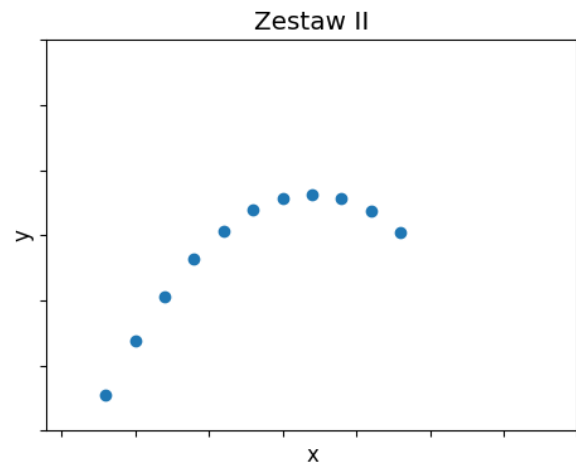
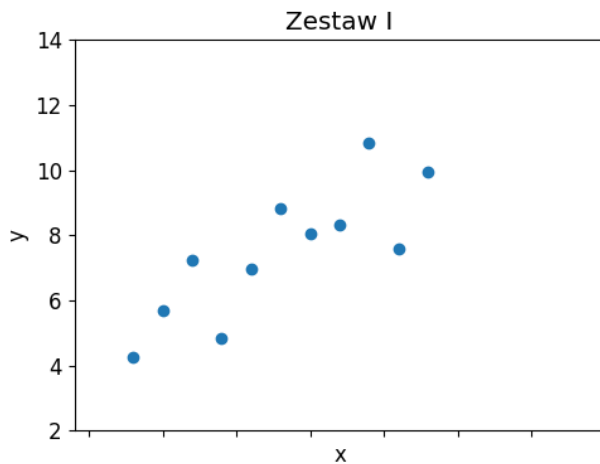
datasets = [
    ('I', df_anscombe['x1'], df_anscombe['y1']),
    ('II', df_anscombe['x2'], df_anscombe['y2']),
    ('III', df_anscombe['x3'], df_anscombe['y3']),
    ('IV', df_anscombe['x4'], df_anscombe['y4'])
]

for i, (title, x_data, y_data) in enumerate(datasets):
    ax = axes[i]
    ax.scatter(x_data, y_data)
    ax.set_title(f'Zestaw {title}')
    ax.set_xlabel('x')
    ax.set_ylabel('y')
    ax.set_xlim(2, 20)
    ax.set_ylim(2, 14)

plt.tight_layout()
plt.show()
```

Wczytane dane Anscombe'a:

	x1	y1	x2	y2	x3	y3	x4	y4
0	10.0	8.04	10.0	9.14	10.0	7.46	8.0	6.58
1	8.0	6.95	8.0	8.14	8.0	6.77	8.0	5.76
2	13.0	7.58	13.0	8.74	13.0	12.74	8.0	7.71
3	9.0	8.81	9.0	8.77	9.0	7.11	8.0	8.84
4	11.0	8.33	11.0	9.26	11.0	7.81	8.0	8.47



Zadanie 2

Przeprowadź eksploracyjną analizę danych (EDA) dla zbioru danych o domach
house_train.csv

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("house_sales.csv", sep="\t")
df["DocumentDate"] = pd.to_datetime(df["DocumentDate"])
```

```

df["Year"] = df["DocumentDate"].dt.year

fig, axes = plt.subplots(2, 2, figsize=(12, 9))
fig.suptitle("EDA – dane o sprzedaży domów w Seattle", fontsize=14,
fontweight="bold")

sns.histplot(data=df, x="SalePrice", bins=50, color="#E63946",
ax=axes[0, 0])
axes[0, 0].set(title="Rozkład ceny sprzedaży", xlabel="Cena
(SalePrice)", ylabel="Liczba transakcji")
axes[0, 0].ticklabel_format(style="plain", axis="x")

sns.scatterplot(data=df, x="SqFtTotLiving", y="SalePrice", alpha=0.15,
s=8, color="#457B9D", ax=axes[0, 1])
axes[0, 1].set(title="Powierzchnia vs Cena", xlabel="Powierzchnia
(SqFt)", ylabel="Cena (SalePrice)")
axes[0, 1].ticklabel_format(style="plain", axis="y")

order = df["PropertyType"].value_counts().index
sns.countplot(data=df, x="PropertyType", hue="PropertyType",
palette=["#2A9D8F", "#F4A261", "#E9C46A"], order=order, legend=False,
ax=axes[1, 0])
axes[1, 0].bar_label(axes[1, 0].containers[0], padding=3)
axes[1, 0].set(title="Liczba nieruchomości wg typu", xlabel="Typ",
ylabel="Liczba")

sns.boxplot(data=df, x="PropertyType", y="SalePrice",
hue="PropertyType", palette=["#2A9D8F", "#F4A261", "#E9C46A"],
order=order, legend=False, ax=axes[1, 1])
axes[1, 1].set(title="Rozkład cen wg typu nieruchomości", xlabel="Typ
nieruchomości", ylabel="Cena (SalePrice)")
axes[1, 1].ticklabel_format(style="plain", axis="y")

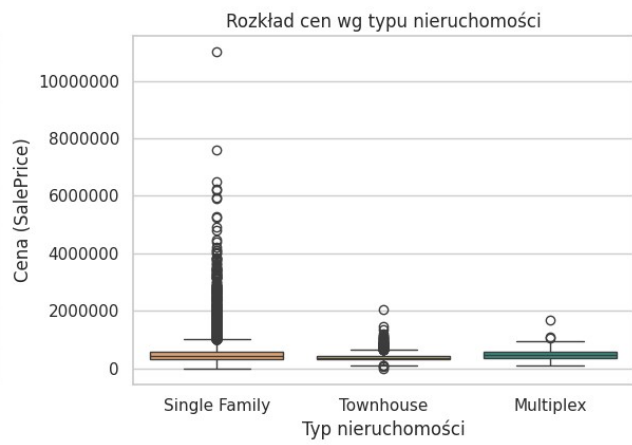
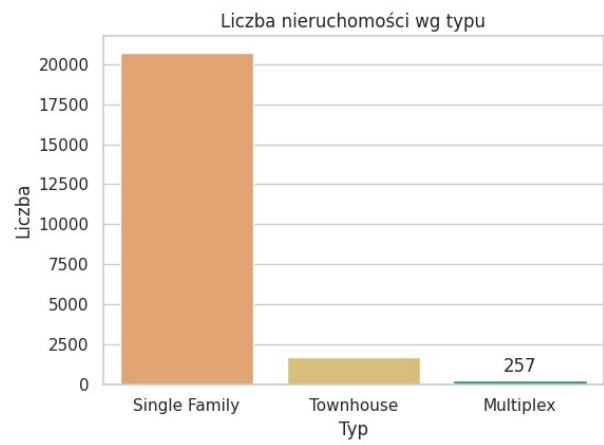
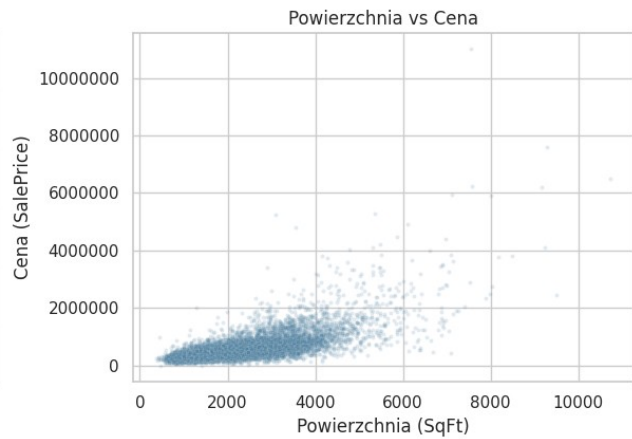
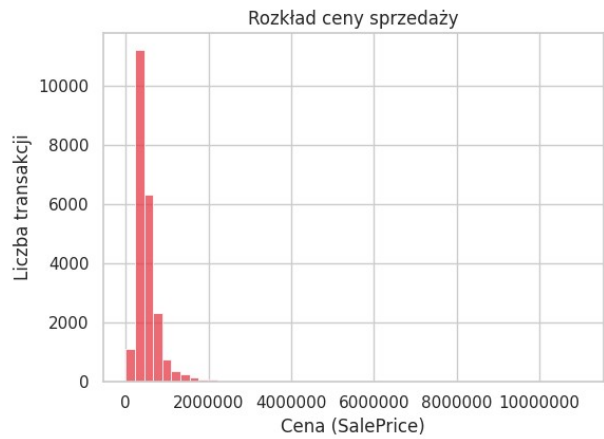
plt.tight_layout()
plt.show()

num_cols = ["SalePrice", "SqFtTotLiving", "SqFtLot", "Bathrooms",
"Bedrooms", "BldgGrade", "YrBuilt", "LandVal", "ImpsVal"]
corr = df[num_cols].corr()

plt.figure(figsize=(9, 7))
sns.heatmap(corr, annot=True, fmt=".2f", cmap="coolwarm", square=True,
linewidths=0.5, annot_kws={"size": 9})
plt.title("Macierz korelacji", fontsize=13, fontweight="bold", pad=12)
plt.tight_layout()
plt.show()

```

EDA - dane o sprzedaży domów w Seattle



Macierz korelacji

